

Assignment-1

CSL7120 (ML-DL-Ops)

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GitHub Link : [GitHub](#)

Colab Link : [🔗 Assignment1_DLOps.ipynb](#)

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Introduction

Objective:

This report details the performance analysis of ResNet-18 and ResNet-50 architectures on the MNIST and FashionMNIST datasets. The experiments evaluate the impact of hyperparameters such as batch size, optimizers, and learning rates on model accuracy and training efficiency. Additionally, a comparison between deep learning models and traditional Support Vector Machine (SVM) classifiers is presented.

Datasets:

Two benchmark datasets were used in this assignment:

1. **MNIST:** Consists of 70,000 grayscale images of handwritten digits (0–9), each of size 28x28 pixels.
2. **FashionMNIST:** Contains 70,000 grayscale images of fashion items (10 classes) with the same dimensions as MNIST.

Models Used:

- **ResNet-18 and ResNet-50:** Deep convolutional neural networks designed for image classification. Both models were trained **from scratch** with randomly initialized weights, without any pre-training.
 - **SVM (Support Vector Machine):** Kernel-based classifiers with **poly** and **rbf** kernels, applied on flattened images to compare classical ML performance with deep learning models.
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Experimental Setup

Data Splits:

- Each dataset was split into **70% training, 10% validation, and 20% testing**.
- For SVM experiments, a **80–20 train-test split** was used, without separate validation.

Hardware:

- Experiments were executed on **Google Colab GPU runtime** (NVIDIA Tesla T4).
- CPU experiments were performed on the same environment by forcing the device to `cpu` in code.

Hyperparameters:

Hyperparameter	Values Tested
Optimizer	SGD, Adam
Learning Rate	0.001, 0.0001
Batch Size	16, 32
Epochs	3 and 5
AMP (Automatic Mixed Precision)	True
Pin Memory	True for GPU, False for CPU

Results Tables

QUES 1. A) PART

MNIST DATASET

			Test Classification Accuracy (in %)	
Batch Size	Optimizers	Learning Rate	ResNet-18	ResNet-50
16	SGD	0.001	96.66666667	95.70833333
16	SGD	0.0001	62.71666667	38.74166667
16	Adam	0.001	98.95833333	98.03333333
16	Adam	0.0001	98.68333333	98.88333333
32	SGD	0.001	87.65	84.1
32	Adam	0.001	98.375	97.3
32	Adam	0.0001	98.75	97.65

FASHION MNIST DATASET

			Test Classification Accuracy (in %)	
Batch Size	Optimizers	Learning Rate	ResNet-18	ResNet-50
16	SGD	0.001	74.5	76.35833333
16	SGD	0.0001	68.56666667	52.00833333
16	Adam	0.001	92.39166667	90.575
16	Adam	0.0001	91.45	90.45833333
32	SGD	0.001	78.65	70.025
32	SGD	0.0001	58.875	38.63333333
32	Adam	0.0001	90.525	88.9

Hyperparameter Variation (Varying Epochs & Pin Memory)

Focusing on the best-performing base configuration (**ResNet-18, Adam, 0.001**), further experiments were conducted by varying the number of epochs and the **pin_memory** setting.

DATASET : MNIST			
USE_AMP=True , Batch Size = 16, Optimizer =Adam, Learning Rate = 0.001			
Epochs	pin_memory	Test Classification Accuracy (in %)	Test Time (ms)
3	TRUE	98.875	27886.6365
3	FALSE	98.69166667	27413.60712
5	TRUE	99.20833333	27207.31425
5	FALSE	98.94166667	26939.31174

Q1(b): SVM Classifier Analysis

A Support Vector Machine (SVM) was trained to provide a baseline comparison against the deep learning models.

Dataset	Kernel	Test Accuracy (%)	Training Time (ms)
MNIST	rbf	94.04%	11690.43
MNIST	poly	90.74%	24753.76
FASHION MNIST	rbf	86.85%	9543.04
FASHION MNIST	poly	83.85%	12336.26

Q2. The following experiments were conducted on the FashionMNIST dataset using a constant **Batch Size of 16**, a **Learning Rate of 0.001**, and **USE_AMP=True**.

Computer	Batch Size	Optimizer	Learning Rate	Test Classification Accuracy (in %)		Train Time (in ms)		FLOPs	
				ResNet-18	ResNet-50	Res Net 18	ResNet-50	Res Net 18	ResNet-50
CPU	16	SGD	0.001	66.92	35.24	747258.58	2388937.35	3,649,611,796	8,260,816,916
CPU	16	Adam	0.001	86.22	88.58	761033.68	2573243.93	3,649,611,796	8,260,816,916
GPU	16	SGD	0.001	80.23	69.53	245974.69	443797.12	3,649,611,796	8,260,816,916
GPU	16	Adam	0.001	88.18	89.86	246755.37	455382.7	3,649,611,796	8,260,816,916

Analysis

ResNet Performance:

- **Observations:** While ResNet-50 is theoretically more powerful due to its 50 layers and bottleneck designs, **ResNet-18** consistently achieved higher or comparable accuracy on the MNIST dataset.
- **Analysis:** For low-resolution (28x28) grayscale images like those in MNIST, the extreme depth of ResNet-50 can lead to overfitting or optimization challenges. ResNet-18

Influence of Optimizers (Adam vs. SGD)

- **Observations: Adam outperformed SGD** in almost every trial. Specifically, at a learning rate of 0.001, Adam reached **~98.9%** accuracy, whereas SGD required careful tuning to cross 96%.
- **Analysis:** Adam's adaptive learning rate mechanism allows it to handle sparse gradients more effectively during the initial phases of training on these datasets. SGD proved to be highly sensitive; at a learning rate of 0.0001, it failed to reach the **80% accuracy threshold**, yielding only **62.7%** for ResNet-18 and an even lower **38.7%** for ResNet-50.

Optimization via Epochs and Pin Memory

Based on the best base configuration (**ResNet-18, Adam, 0.001**), further tuning was performed:

- **Accuracy:** Increasing training to **5 epochs** pushed accuracy to its peak of **99.21%**.
- **Efficiency:** Enabling `pin_memory=True` improved data transfer speed between the CPU and GPU, reducing the total training time to **27,207 ms**.

SVM Experiments Summary

- Kernels tested: `poly` and `rbf`.
- A Support Vector Machine (SVM) was trained to provide a baseline comparison against the deep learning models.
- Both MNIST and FashionMNIST datasets were used.

Observations:

Kernel Performance and Accuracy

- **Observations:** The **RBF kernel** was superior to the Polynomial kernel across both datasets. On MNIST, RBF reached **94.04%**, while Fashion-MNIST, being a more complex dataset (textural clothing patterns vs. simple digits), saw lower performance at **86.85%**.
- **Analysis:** Traditional SVMs lack the spatial hierarchy awareness that CNNs possess. While they perform reasonably well on flattened vectors for simple digits, they struggle with the intricate spatial features of Fashion-MNIST.

Computational Efficiency

- **Training Time:** SVM training time scales poorly with dataset size. As the training set grows, the quadratic complexity of SVM training makes it significantly slower than the batch-processed CNNs.
 - **Comparison:** Even though SVM training on MNIST was fast (~11 seconds), it does not scale to the millions of parameters handled efficiently by ResNets.
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CPU vs GPU Comparison

Comparative Analysis: Training Time

- **Hardware Acceleration:** Transitioning to a GPU backend provided a massive reduction in training latency; for ResNet-50 with Adam, the GPU was **~5.6x faster** than the CPU, slashing training from **42.9 minutes to 7.6 minutes**.
- **Architecture Latency:** ResNet-50 consistently demanded more training time than ResNet-18. On a CPU, ResNet-50 was **over 3x slower**, highlighting the heavy computational burden of deep bottleneck architectures when parallelization is limited.

Computational Complexity (FLOPs)

- **Resource Demand:** ResNet-50 utilizes **8,260,816,916 FLOPs**, which is **2.26x higher** than the **3,649,611,796 FLOPs** required by ResNet-18.
- **Efficiency Trade-offs:** While ResNet-50 reached the highest accuracy (**89.86%**), ResNet-18 offers a more efficient profile for FashionMNIST, achieving **88.18% accuracy** using less than half the total FLOPs.

Accuracy and Optimizer Stability

- **Optimizer Performance:** Adam significantly outperformed SGD; while all Adam GPU runs exceeded the **80% accuracy threshold**, the ResNet-50/SGD configuration on CPU failed with only **35.24% accuracy**.
 - **Model Depth Gain:** The increased complexity of FashionMNIST saw a slight performance gain from ResNet-50's deeper feature extraction compared to its performance on simple MNIST digits.
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